

A Packing of 18 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 18 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.540834. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.532951, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 18 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.540834 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 18$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.532951, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 18/17 = 2.541176$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N18.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
158802184928489146239159665217755527389525785880890376815869787741783727
788902168632436533935339974981069894417090539660938052911121216431158997
500388867389350425906752542846468209209577646823
```

denominator:

```
625000000001855251979654227144534174540439435481835787116133774318409443
143198582193254493844913828186759040516122689001466534547010859237587617
90121017623651166607561530750530000150346566712
```

B Coordinates

18 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N18.txt`.

```
-85593453256 388503353959 -164274609262 49064505709200982623422 378752265084 -105411461437 -1212662301285130468652 433495958870 493363647737 2087920172
316474022240 -316973534603 -179647091718 5948125806292312732366 -183689909152 -262564242907 59126858563279735956 -124078734944 343627716844 606354556474
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